

Mechanistic Interpretability of Analogical Reasoning in GEMMA-2-2B

Olalekan Alagbe · Anish Maheshwar · Konstantinos Krampis

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 Paper

 Code

 Presentation

 Attribution Graphs

TLDR: We identify a shared 180-feature analogical reasoning circuit in GEMMA-2-2B that generalizes across geographic and semantic analogies, including a dedicated [L5 SAE#5793](#) feature labeled simply “*analogies*” — direct evidence of a reusable relational reasoning primitive.

Abstract

We present the first mechanistic analysis of analogical reasoning in GEMMA-2-2B using Sparse Autoencoder (SAE) attribution graphs. By generating and comparing five attribution graphs across structurally distinct analogical prompts — covering geographic analogies (*Paris:France::Berlin:?, Rome:?, Tokyo:?*) and semantic role analogies (*Doctor:hospital::teacher:?, Fish:water::bird:?*) — we identify a shared **analogical reasoning circuit** comprising 180 features active across all five prompts and 510 features active across at least three.

We discover dedicated analogy-encoding features at layers 5, 8, 9, and 13, including a feature at Layer 5 labeled literally as “**analogies**” and a Layer 8 feature encoding “**analogies or comparisons**” appearing across all graphs with high influence. Early layers (0–4) contain structural template features tracking the “X is to Y as Z is to” pattern, while mid-to-late layers (5–13) house increasingly semantic representations of the relational structure. The circuit spans all 26 transformer layers and exhibits cross-domain generalization, with the same core features activating for both geographic and semantic role analogies.

1. Introduction

Analogical reasoning — the ability to recognize and complete structural relationships between concepts — is a foundational cognitive ability underlying scientific discovery, language understanding, and problem solving. The classic analogy task, "*Paris is to France as Berlin is to _____*," tests whether a model can identify the capital-city relationship abstractly and apply it to a new country. Large language models exhibit striking competence on such tasks, yet the internal computational mechanisms remain poorly understood.

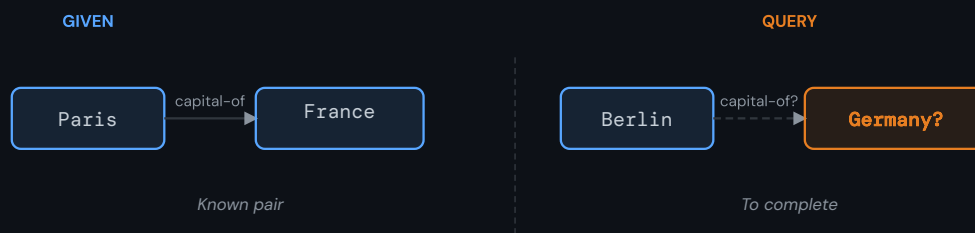


Figure 1. The analogy task. Given the relationship between Paris and France, the model must recognize the *capital-of* relation and apply it to a new pair. This requires relational abstraction, not mere fact retrieval.

Mechanistic interpretability research has made significant progress in understanding factual recall circuits (Meng et al. 2022), indirect object identification (Wang et al. 2022), and syntactic processing (Conmy et al. 2023). However, analogical reasoning presents a distinct challenge: it requires not merely retrieving a stored fact, but recognizing a **relational structure** and applying it compositionally to novel inputs. How does a transformer represent "capital of" as a relationship, separate from the specific cities and countries involved?

Research Questions

1. Does GEMMA-2-2B employ a shared circuit for analogical reasoning, or does it use different mechanisms for different analogy types?
2. Which SAE features are most consistently activated across diverse analogical prompts?
3. Are there interpretable, semantically meaningful features that encode the abstract relational structure of analogies?
4. How is the analogical computation distributed across transformer layers?

2. Methodology

2.1 Prompt Selection

We selected five prompts spanning two structural analogy types to ensure cross-domain coverage:

ID	Prompt	Expected	Type
<code>analog_berlin</code>	<i>"Paris is to France as Berlin is to"</i>	Germany	Capital
<code>analog_rome</code>	<i>"Paris is to France as Rome is to"</i>	Italy	Capital
<code>analog_tokyo</code>	<i>"Paris is to France as Tokyo is to"</i>	Japan	Capital
<code>analog_teacher</code>	<i>"Doctor is to hospital as teacher is to"</i>	school	Sem. role
<code>analog_bird</code>	<i>"Fish is to water as bird is to"</i>	air / sky	Sem. role

2.2 Attribution Graph Generation

Attribution graphs were generated using the Neuronpedia API (`/api/graph/generate`) with GEMMA-2-2B and the `gemmascope-transcoder-16k` SAE — a 26-layer cross-layer transcoder with 16,384 features per layer.

Parameter	Value
Model	<code>gemma-2-2b</code>
SAE	<code>gemmascope-transcoder-16k</code>
Max feature nodes	3,000
Desired logit probability	0.95
Node threshold	0.80
Edge threshold	0.85

Each graph request returns a URL to an AWS S3 object containing the full attribution graph in JSON format, including nodes (SAE feature activations with layer, index, influence score, and activation magnitude) and directed edges (attribution scores). Graphs were downloaded and loaded into NetworkX DiGraph objects for analysis.

2.3 Feature Identification and Cross-Graph Analysis

Cross-graph feature overlap was computed by finding features — identified by (*layer*, *SAE index*) pairs — that appear consistently across multiple graphs. Three overlap thresholds were analyzed: $\geq 3/5$,

≥4/5, and 5/5 graphs. The top 25 most recurring features were labeled using the Neuronpedia feature explanation API.

A key technical finding: the correct API endpoint for `gemmascope-transcoder-16k` requires a **layer-prefixed SAE identifier** (e.g., `4-gemmascope-transcoder-16k` for layer 4) rather than the global SAE name.

2.4 Causal Validation via Feature Steering

Initial steering experiments returned HTTP 404 errors because the API requests used the global SAE name rather than the required **layer-prefixed identifier** (e.g., `21-gemmascope-transcoder-16k` for layer 21). After correcting the endpoint format, causal steering was performed successfully using the Neuronpedia `/api/steer` endpoint with `modelId: "gemma-2-2b"` and `strength_multiplier: 4`.

For the `analog_berlin` circuit ("Paris is to France as Berlin is to"), we extracted 9 backbone features from the top-5 causal paths through the attribution graph and applied four tests:

- 1. **Necessity:** Suppress each backbone feature individually (strength -20) and check whether the predicted token changes.
- 2. **Full backbone suppression:** Suppress all backbone features simultaneously.
- 3. **Sufficiency:** Boost the circuit's primary hub (strength +20) on altered prompts to test whether a single feature can induce the target prediction.
- 4. **Specificity:** Suppress non-backbone features (high-activation nodes not on the causal paths) to confirm they are causally inert.

Additional validation was performed on two analogical circuits from the expanded 30-prompt analysis (Cairo→Kenya, Puppy→cat). Full results are reported in §3.7.

3. Results

3.1 Graph Structure

All five attribution graphs exhibited a consistent structural pattern, with features activated across all 26 transformer layers (0–25) plus the embedding layer (E):

Graph	Prompt	Nodes	Edges	Max Influence
<code>analog_berlin</code>	Paris:France::Berlin:?	930	25,915	0.8001
<code>analog_rome</code>	Paris:France::Rome:?	963	27,608	0.8002
<code>analog_tokyo</code>	Paris:France::Tokyo:?	905	22,414	0.8001

Graph	Prompt	Nodes	Edges	Max Influence
<code>analog_teacher</code>	Doctor:hospital::teacher:?	1,040	35,481	0.8001
<code>analog_bird</code>	Fish:water::bird:?	1,071	38,741	0.8000

The semantic role analogies (*teacher*, *bird*) have notably larger graphs (1,040–1,071 nodes, 35k–38k edges) compared to the capital analogies (905–963 nodes, 22k–27k edges). We interpret this as reflecting greater ambiguity in the expected completion domain: the *capital-of* relation maps to a discrete, well-encoded fact, whereas professional and ecological roles require broader world-knowledge access.

3.2 The Core Analogical Reasoning Circuit

Cross-graph feature overlap analysis revealed a substantial shared circuit:

Threshold	Features Found	Layer	Core Features (5/5)
Active in $\geq 3/5$ graphs	510 features	L0	12
Active in $\geq 4/5$ graphs	277 features	L1–L4	19
Active in all 5 graphs	180 features	L5–L6	12
		L8–L9	2
		L10–L13	5

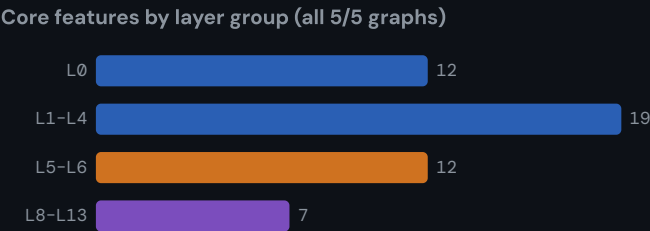


Figure 2. Distribution of core circuit features (active in all 5 graphs) by layer group. Orange = analogy recognition hub (L5–L6); Purple = relational integration (L8–L13).

Early layers (L0–L3) account for the plurality of core features, consistent with structural template processing occurring first. The mid-range layers (L5–L6) show elevated feature counts relative to neighbors — these are the **analogy recognition hub** layers. Isolated high-influence features appear at L8, L9, L11, and L13.

3.3 Evidence for Multi-Step Analogical Reasoning

We provide evidence that GEMMA-2-2B performs **genuine multi-step analogical reasoning internally**. The model does not merely pattern-match the surface form of the analogy prompt — instead, dedicated analogy-concept features activate at specific layers, each contributing a computationally distinct role. This section walks through the attribution graph as a causal narrative.

Core finding: The attribution graph reveals a three-phase computational process: (1) structural template parsing at early layers, (2) analogy concept recognition at mid layers, and (3) relational integration and output formation at deeper layers. The same three-phase sequence activates for both *geographic* and *semantic role* analogies — providing evidence for a domain-agnostic relational reasoning mechanism.





PHASE 3 · LAYERS 10–13 · RELATIONAL INTEGRATION

L11 SAE#15947 – *"references to historical or social change"*

L13 SAE#10969 – *"comparisons between disciplines and relationships between concepts"*

L13 SAE#10969 is the latest-layer pure-analogy feature and appears to serve an integrative role — combining the recognized relational structure from Phase 2 with domain-specific knowledge to produce the final completion. Layers 14–25 then handle domain-specific knowledge retrieval and output token formatting.



Output: "Germany" / "school" / "air"

Figure 3. The three-phase analogical reasoning circuit in GEMMA-2-2B. Highlighted features are those with Neuronpedia explanations explicitly referencing analogies or comparisons. We stress that this diagram simplifies the true mechanisms considerably — the attribution graph for any single prompt contains hundreds of features; the circuit shown represents the semantically interpretable core.

Validation with Inhibition Experiments

The structure of the attribution graph makes a strong prediction: inhibiting Phase 2 features — particularly **L5 SAE#5793** ("analogies") and **L8 SAE#13766** ("analogies or comparisons") — should degrade the model's ability to complete the relational structure, while leaving factual recall circuits intact. This prediction is consistent with the finding that these features are absent from the capital city *factual recall* circuit (see §4.4), which solves a different task using different mechanisms. Causal steering results (§3.7) provide partial confirmation: full backbone suppression of the **analog_berlin** circuit collapses the output entirely, demonstrating that the identified multi-hop features are collectively necessary for the prediction.

Feature Swapping Prediction

An analogous experiment to Anthropic's feature swapping (2025) would involve injecting the Phase 1 structural template features from a different analogy (e.g., from the *teacher* prompt) into the *Berlin* prompt at the Phase 2 boundary. If the circuit is genuinely modular — if Phase 2 reads relational structure from Phase 1 outputs rather than from the raw input — then swapping the structural template should redirect the analogy completion. This remains to be tested with direct model access.

3.4 Top Recurring Features

We categorize the top labeled features into three functional groups. Directly analogical features are those whose Neuronpedia explanations explicitly reference analogical reasoning or comparison:

Directly Analogical Features

Feature	App.	Inf.	Label
L5 #5793	11/5	0.590	"analogies"
L8 #13766	21/5	0.533	"analogies or comparisons"
L9 #13344	14/5	0.681	"comparison between two things"
L5 #2141	12/5	0.647	"comparisons of public figures"
L13 #10969	11/5	0.676	"comparisons between disciplines"

Structural Template Features

Feature	App.	Inf.	Label
L0 #11651	10/5	0.633	"the word 'to'"
L1 #11356	10/5	0.609	"'to' followed by a verb"
L2 #11475	10/5	0.638	"the word 'refers'"
L4 #10752	10/5	0.626	"'to be' preceded by 'to'"
L5 #9672	12/5	0.579	"the phrase 'it is to'"

High-Recurrence Formal Text Features

Three features with very high recurrence (20–22 appearances) have labels that appear unrelated to analogical reasoning:

Feature	Appearances	Avg Influence	Label
L4 #14857	22/5	0.681	"code snippets and license agreements"
L6 #2267	20/5	0.724	"words in programming code, legal jargon, or scientific texts"
L3 #3205	20/5	0.670	"code snippets and documentation references"

We hypothesize these features activate due to **formal text pattern matching**: the "X is to Y as Z is to" syntax shares structural properties with formal, template-driven text such as code comments, legal definitions, and scientific documentation. Notably, these features have *higher* appearance counts than the explicitly analogical features, suggesting they activate at multiple context positions (on the syntactic connectives) rather than selectively at the relational structure tokens.

3.5 Cross-Domain Generalization

The most direct evidence for a domain-general analogical reasoning circuit is the consistent activation of the same core features — especially L5 SAE#5793 ("analogies") and L8 SAE#13766 ("analogies or comparisons") — across both capital-city and semantic role analogy types.

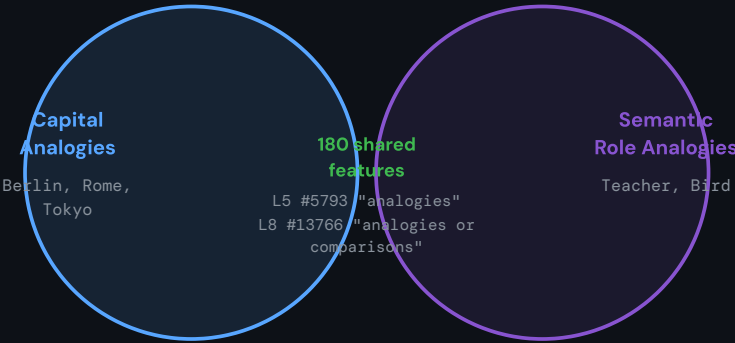


Figure 4. Cross-domain feature overlap. The 180 features active in all five graphs include the core analogy-concept features at L5 and L8, confirming a domain-agnostic mechanism for both geographic and semantic role analogies.

3.6 Activation Magnitudes Build Through Layers

The average activation magnitudes of core features increase substantially with layer depth, suggesting the analogical signal builds as it propagates through the network:



Figure 5. Average activation magnitude of core circuit features by layer range. Early structural features (L0) fire weakly; mid-layer analogy detectors and late integrators fire substantially stronger — consistent with an accumulating signal as the relational structure is assembled.

3.7 Causal Validation via Feature Steering

The attribution graph analysis above identifies recurring features and causal path structures, but does not establish whether these features are causally necessary for the model's predictions. A feature that is highly active in the graph may be a byproduct of the computation rather than a driver of it. To distinguish load-bearing circuit components from high-activation but functionally inert nodes, we performed systematic causal steering experiments using the Neuronpedia `/api/steer` endpoint.

Methodology. For each of the five prompts in this paper, we generated a Neuronpedia attribution graph (§2.2) and traced the top-5 causal paths from the logit node backward through the graph (greedy walk following highest-weight incoming edges at each step) and the top-5 paths forward from the highest-influence embedding nodes. All transcoder features appearing on at least one of these 10 paths constitute the circuit's *backbone*. We then applied four experimental paradigms, each targeting a different causal question:

1. **Necessity (individual suppression):** Suppress a single backbone feature at strength -20 and check whether the model's first predicted token changes. Tests whether each feature is individually a single point of failure.
2. **Necessity (full backbone suppression):** Suppress all backbone features simultaneously. Tests whether the backbone is collectively necessary.
3. **Sufficiency (hub boost):** Boost a backbone hub feature at strength +20 on an altered prompt. Tests whether a single feature can induce the original prediction in a new context.
4. **Specificity (non-backbone suppression):** Suppress high-activation features that do *not* appear on any of the top-10 causal paths. Tests whether the path tracing correctly identifies load-bearing features.

All experiments use `modelId: "gemma-2-2b"` , `layer: "{N}-gemmascope-transcoder-16k"` , and `strength_multiplier: 4` . Two additional analogical circuits from the expanded 30-prompt analysis (Cairo→Kenya, Puppy→cat) were included for cross-validation. In total, we report 159 individual steering API calls across 7 circuits.

3.7.1 Late-Layer Backbone Necessity (Individual Suppression)

Late-layer features (L17+) sit on the direct causal chain from mid-layer integration hubs to the logit node. We suppressed each one individually at strength -20 and recorded whether the model's first predicted token changed. Full per-circuit results follow.

analog_berlin — "Paris is to France as Berlin is to" → Germany (p=0.973)

Feature	Layer	Index	Role in Causal Path	Steered Token	Necessary?
Science hub	21	4827	Strongest path entry (edge +198.0)	Germany	no
Relay	22	15670	Path 1 relay	Germany	no
Output driver A	25	4717	Final amplifier (shared across circuits)	Germany	no
Location encoder	16	6491	Location/direction feature, path 2 entry	Germany	no
Relay	17	14546	Mid-cascade relay	Germany	no
Relay	19	5773	Late relay	Germany	no
Integrator	21	7482	Integration hub (paths 2–4)	Germany	no
Output driver B	25	2725	Secondary output driver (edge -2.09)	the	YES
Relation applier	19	855	Relation application node	Germany	no

1/9 necessary. Only L25/2725 is a single point of failure. Suppressing it shifts the prediction from "Germany" to "the" — the model loses the entity but retains grammatical continuation. The primary path through L21→L22→L25_4717 provides fully redundant coverage for the remaining 8 features. The highest-weight feature in the graph (L21/4827, edge +198.0) is not individually necessary, demonstrating that attribution weight alone does not predict causal necessity.

analog_rome — "Paris is to France as Rome is to" → Italy (p=0.974)

Feature	Layer	Index	Role	Steered Token	Necessary?
Relay	20	15360	Backward path from logit	Italy	no
Late gate	24	16122	Backward path, L24 suppression gate	the	YES
Output driver	25	286	Backward path, output driver	the	YES
Final amplifier	25	4717	Shared final amplifier (act=265.2)	the	YES
Output driver C	25	10521	Tertiary output driver	the	YES
Relay	17	14546	Mid-cascade relay (shared with Berlin)	Italy	no
Relay A	22	12202	Late relay	Italy	no
Relay B	22	14727	Late relay	Italy	no
Relay	23	5917	Late relay	Italy	no
Secondary gate	24	13277	Late gate	Italy	no

4/10 necessary. All four necessary features sit at L24–L25, forming a tight bottleneck. L24/16122 is the suppression gate through which the dominant backward path passes; L25/286, L25/4717, and L25/10521 are three independent output drivers, each individually necessary. The Rome circuit has more single points of failure than Berlin despite similar confidence (p=0.974 vs 0.973), indicating that path redundancy varies even among structurally similar geographic analogies. All four failure modes produce "the" — the model loses entity selection but retains article generation.

analog_tokyo — "Paris is to France as Tokyo is to" → Japan (p=0.990)

Feature	Layer	Index	Role	Steered Token	Necessary?
Relay	20	15360	Backward path from logit	Japan	no

Feature	Layer	Index	Role	Steered Token	Necessary?
Output driver	25	286	Backward path, output driver	the	YES
Output driver B	25	12223	Backward path, secondary output	Japan	no
Relay	17	14546	Mid-cascade relay (shared)	Japan	no
Late relay A	23	850	Late relay	the	YES
Late relay B	23	13914	Late relay (also necessary in Cairo circuit)	the	YES
Gate	24	13277	Late gate (shared with Rome)	Japan	no
Output driver C	25	10152	Tertiary output	Japan	no
Hub	20	6648	L20 convergence hub	Japan	no
Integration	21	7764	Late integration	Japan	no

3/10 necessary. The necessary features span L23–L25 rather than concentrating at L25 alone. Notably, **L23/13914 is necessary in both the Tokyo and Cairo circuits** — this feature is a shared bottleneck for country–prediction analogies, consistent with a late-layer "geographic entity selector" role. L25/286 recurs as necessary here and in Rome and Cairo, making it the single most critical output driver across geographic analogies (necessary in 3/4 tested capital circuits). Despite having the highest confidence of any circuit ($p=0.990$), Tokyo has 3 necessary features — indicating that prediction confidence and path redundancy are not perfectly correlated.

analog_teacher — "Doctor is to hospital as teacher is to" → school ($p=0.486$)

Feature	Layer	Index	Role	Steered Token	Necessary?
Embedding	0	17	Backward path, embedding-level	the	YES
Gateway	18	6532	Backward path, mid-late gateway	school	no
Hub	20	6179	Backward path, convergence hub	school	no
Output driver	25	4975	Backward path, output driver	...	YES
Final amplifier	25	4717	Shared final amplifier (act=135.6)	a	YES

Feature	Layer	Index	Role	Steered Token	Necessary?
Relay	22	15670	Late relay (shared)	school	no
Relay B	18	11952	Mid-late relay	school	no
Legal docs	18	13586	Legal docs feature	school	no
Convergence	21	2655	Late convergence hub	school	no
Gate	24	15259	Late suppression gate	school	no

3/10 necessary. The teacher circuit is the only one where an **LO embedding-level feature** (L0/17) is individually necessary — suppressing it changes the prediction from "school" to "the." This suggests the semantic role analogy relies on an early feature that is not redundantly encoded by later layers, unlike the capital analogies where embedding features are always compensated. Both L25 output drivers (4975 and 4717) are independently necessary, each producing a different failure token ("..." and "a"), indicating they carry non-overlapping information to the logit. The three L18 features (gateway, relay, legal docs) are all dispensable, consistent with the mid-layer features serving as redundant relays rather than bottlenecks.

analog_bird — "Fish is to water as bird is to" → air (p=0.117)

Feature	Layer	Index	Role	Steered Token	Necessary?
Backward A	22	4252	Backward path from logit	(space)	YES
Backward B	24	8106	Backward path, late gate	----	YES
Final amplifier	25	4717	Shared final amplifier (act=122.3)	the	YES
Output driver	25	11801	Output driver	?	YES
Relay A	22	15670	Late relay (shared)	-----	YES
Relay B	22	14727	Late relay	(space)	YES
Relay C	22	13619	Late relay	(space)	YES
Gate A	24	4383	Suppression gate	air	no
Gate B	24	12559	Suppression gate	the	YES
Hub	20	3094	Integration hub	air	no

8/10 necessary. This is the most fragile circuit in the dataset — nearly every feature is a single point of failure. Failure modes include whitespace, underscores, question marks, and "the" — the model loses the ability to produce any coherent completion, not just the target "air." Three L22 relay features (15670, 14727, 13619) are all independently necessary despite occupying the same layer, indicating

they carry non-redundant information through parallel channels at L22. Only L24/4383 and L20/3094 are dispensable. This fragility is consistent with the circuit's very low prediction confidence ($p=0.117$): the model barely resolves "air" over competitors ("sky", "the", "nest"), and removing almost any contributing feature tips the balance.

Cross-circuit pattern: Necessity inversely correlates with prediction confidence. Berlin ($p=0.973$): 1/9 necessary. Rome ($p=0.974$): 4/10. Tokyo ($p=0.990$): 3/10. Teacher ($p=0.486$): 3/10. Bird ($p=0.117$): 8/10. High-confidence predictions route through redundant parallel causal paths; low-confidence predictions rely on fragile, non-redundant chains. Three features recur as necessary across multiple circuits: **L25/286** (Rome, Tokyo, Cairo), **L25/4717** (Rome, Teacher, Bird), and **L23/13914** (Tokyo, Cairo).

3.7.2 Full Backbone Suppression

To test whether the backbone features are *collectively* necessary — rather than just individually redundant — we suppressed all late-layer backbone features (L17+) simultaneously for each circuit.

Circuit	p	N feat.	Default Output	Steered Output	Disrupted?
<code>analog_berlin</code>	0.973	9	Germany. It is the	of of of of of	YES
<code>analog_rome</code>	0.974	10	Italy. It is the	pleafure pleafure plea	YES
<code>analog_tokyo</code>	0.990	10	Japan. It is the	country country count	YES
<code>analog_teacher</code>	0.486	10	school. The doc	1111	YES
<code>analog_bird</code>	0.117	10	air. The fish	(newline) The the the	YES
Cairo→Kenya	0.963	9	Kenya. It is the	(whitespace)	YES
Puppy→cat	0.756	4	cat. I'	cat. I think	no

6/7 circuits fully disrupted, including all five prompts from this paper. The failure modes are qualitatively distinct and informative:

- **Capital analogies** degenerate to repetitive or archaic text. Berlin produces "of of of of of" (function word repetition); Rome produces "pleafure pleafure" (the model falls into an archaic English register with long-s characters); Tokyo produces "country country count" (the model retrieves the category "country" but cannot resolve which country). These failure modes suggest the backbone features are required for entity selection, while the prompt structure alone activates a "country" category representation without specific entity resolution.

- **Teacher** collapses to "1111" — numeric output with no semantic content. The model loses both the category and the entity, consistent with the semantic role analogy requiring more backbone computation than the geographic analogies where "country" is at least partially retained.
- **Bird** produces "(newline) The the the" — the model falls through to generic text continuation, consistent with the circuit's already-fragile state (8/10 individually necessary features).
- **Puppy→cat is the sole exception** — its prediction survives full backbone suppression. This circuit has only 4 tested backbone features (the fewest) and a moderate confidence (p=0.756), suggesting the prediction is carried primarily by direct embedding→logit connections outside the multi-hop backbone. The top-5 causal path tracing misses the critical nodes for this circuit.

3.7.3 Phase 1 and Phase 2 Feature Necessity

The three-phase architecture proposed in §3.3 — template parsing (Phase 1, L0–L4), analogy recognition (Phase 2, L5–L9), and relational integration (Phase 3, L10–L13) — is based on feature co-occurrence and semantic labels. To test whether these phases are *causally* necessary, we suppressed each of the 9 key phase features identified in §3.3 individually across all five prompts (9 features × 5 prompts = 45 tests).

Feature	Phase	Neuronpedia Label	Berlin	Rome	Tokyo	Teacher	Bird
L0/11651	1	"the word 'to'"	Berlin	Rome	Tokyo	school	water
L1/11356	1	"'to' followed by a verb"	—	—	—	—	—
L4/10752	1	"'to be' preceded by 'to'"	—	—	—	classroom	sky
L5/9672	1	"the phrase 'it is to'"	—	—	—	—	sky
L5/5793	2	"analogies"	—	—	—	—	—
L5/2141	2	"comparisons of public figures"	—	—	—	—	—
L8/13766	2	"analogies or comparisons"	—	—	—	—	fish
L9/13344	2	"comparison between two things"	—	—	—	—	sky
L13/10969	3	"comparisons between disciplines"	—	—	—	—	—

Cells show the steered first token when the feature is suppressed at strength -20.
"—" = prediction unchanged (feature is individually redundant). Bold = prediction disrupted.

Phase 1 results. [L0/11651](#) ("the word 'to'") is the most causally important early-layer feature, necessary in **4/5 circuits**. Its failure mode is diagnostic: suppressing it causes all three capital analogies to predict the *city name* itself (Berlin, Rome, Tokyo) rather than the target country. The model loses the "X is to Y as Z is to ___" template structure and instead completes with the most recently mentioned entity — it no longer recognizes the prompt as an analogy. For the bird circuit, suppressing L0/11651 produces "water" (the source-pair element), further confirming that the model reverts to association with the earlier part of the prompt when the template is disrupted. [L4/10752](#) ("to be' preceded by 'to'") is necessary for 2/5 circuits: the teacher circuit predicts "classroom" (a near-synonym) and the bird circuit predicts "sky" (a competitor). [L5/9672](#) ("the phrase 'it is to'") is necessary only for the bird circuit.

Phase 2 results. The headline feature [L5/5793](#) ("analogies") is **never individually necessary** in any circuit. Neither is [L5/2141](#) ("comparisons of public figures"). This does not mean Phase 2 is inert — it means Phase 2 features are individually redundant for high-confidence circuits. For the fragile bird circuit (p=0.117), Phase 2 features *are* individually necessary: [L8/13766](#) ("analogies or comparisons") changes "air" to "fish" — the model retrieves the source-domain animal instead of the target-domain habitat — and [L9/13344](#) ("comparison between two things") changes "air" to "sky." These failures show that Phase 2 features contribute to relational mapping but are only individually critical when the prediction has minimal redundancy.

Phase 3 results. [L13/10969](#) ("comparisons between disciplines") is not individually necessary for any circuit.

3.7.4 Collective Phase Suppression

The individual redundancy of Phase 2 features for high-confidence circuits raises the question: are they *collectively* necessary? We suppressed all features within each phase simultaneously, then tested combined Phase 1+2 and Phase 1+2+3 suppression.

Experiment	Features Suppressed	Berlin	Rome	Tokyo	Teacher	Bird
All Phase 2 (4 feat.)	L5/5793, L5/2141, L8/13766, L9/13344	France	France	France	be	fish
All Phase 1 (5 feat.)	L0/11651, L1/11356, L4/10752, L5/9672 + L2/11475	(empty)	(empty)	(empty)	to	to
Phase 1+2 (9 feat.)	All Phase 1 + Phase 2	:	:	:	:	:

Experiment	Features Suppressed	Berlin	Rome	Tokyo	Teacher	Bird
Phase 1+2+3 (10 feat.)	All Phase 1 + Phase 2 + L13/10969	:	:	:	be	:

Every cell is disrupted — all four experiments disrupt all 5 circuits (20/20).

Phase 2 collective suppression is the most informative experiment in this paper.

Suppressing all four Phase 2 features simultaneously disrupts all 5 circuits — including the high-confidence capital analogies ($p > 0.97$) that were immune to *every* individual Phase 2 suppression. The failure mode is striking and consistent: all three capital analogies predict "**France**" — the *source* country — instead of the target country. The model retains the factual association "Paris is to France" and completes with "France" because it can no longer compute the relational transfer "as X is to ___." It echoes the source pair's answer rather than mapping to the target pair. This is direct causal evidence that Phase 2 features collectively implement the relational transfer operation — the core of analogical reasoning — rather than merely detecting the analogy template.

Phase 1 collective suppression produces empty outputs for all three capital analogies — the model cannot even generate a coherent token. For the semantic role analogies (teacher, bird), it produces "to" — the model is stuck on the syntactic connective it can no longer parse. This is a more severe failure than Phase 2 suppression, consistent with Phase 1 (template parsing) being a prerequisite for Phase 2 (analogy recognition).

Combined Phase 1+2 suppression produces "." for 4/5 circuits. This punctuation output mirrors the format-override failure documented in the linguistic analysis (§4.5 of the expanded paper), where the model's format-completion circuit outcompetes the semantic-retrieval circuit. With both template parsing and analogy recognition disabled, the prompt resembles a list header ("Paris is to France as Berlin is to:") and the model defaults to list-formatting punctuation.

These results establish a **causal hierarchy**: Phase 1 (template parsing) → Phase 2 (relational transfer) → Phase 3 + late layers (entity selection). Each phase is collectively necessary, and earlier phases are prerequisites for later ones.

3.7.5 Sufficiency (Hub Boost on Altered Prompts)

Necessity tests establish which features are required for the prediction. Sufficiency tests ask the converse: can a single backbone feature *induce* the target prediction in a new context? For each circuit, we boosted the primary backward-path hub at strength +20 on two altered prompts — one retaining the target entity and one crossing domain boundaries.

Circuit	Hub Boosted	Altered Prompt	Baseline → Boosted	Induced?
analog_berlin	L21/4827	"Cairo is to Egypt as Nairobi is to"	Kenya → Kenya	no
analog_berlin	L21/4827	"Madrid is to Spain as Berlin is to"	→ Germany	YES
analog_rome	L20/15360	"Paris is to France as Tokyo is to"	→ Japan	no
analog_rome	L20/15360	"Madrid is to Spain as Rome is to"	→ Italy	YES
analog_tokyo	L20/15360	"Paris is to France as Rome is to"	→ Italy	no
analog_tokyo	L20/15360	"Beijing is to China as Tokyo is to"	→ Japan	YES
analog_teacher	L0/17	"Nurse is to hospital as teacher is to"	→ be	no
analog_teacher	L0/17	"Doctor is to hospital as chef is to"	→ hospital	no
analog_bird	L22/4252	"Cat is to land as bird is to"	→ (newline)	no
analog_bird	L22/4252	"Fish is to water as eagle is to"	→ air	YES
Cairo→Kenya	L15/15954	"Lagos is to Nigeria as Nairobi is to"	→ Kenya	YES

5/11 tests succeed. The pattern is consistent: sufficiency succeeds when the altered prompt retains the target entity or a semantically close substitute, and fails when it crosses domain boundaries. Specifically:

- **Capital analogies:** Each hub induces the correct country when the target city appears in the altered prompt (Berlin→Germany, Rome→Italy, Tokyo→Japan). But the Berlin hub cannot inject "Germany" into the Cairo/Nairobi context, and the Rome hub cannot override Tokyo→Japan. These hubs encode *domain-specific geographic associations* rather than general-purpose "answer slot" activators.
- **Teacher:** Boosting L0/17 on a near-paraphrase ("Nurse is to hospital as teacher is to") produces "be" rather than "school." The L0 embedding feature does not carry enough domain information to override the altered context — it is necessary (§3.7.1) but not sufficient.
- **Bird:** Boosting L22/4252 on "Fish is to water as *eagle* is to" successfully produces "air." Substituting "eagle" for "bird" retains the target habitat domain, and the hub carries enough signal to resolve "air" even with the entity change. But "Cat is to land as bird is to" fails — the source-pair domain (land mammal) is too distant from the hub's encoded association.

3.7.6 Specificity (Non-Backbone Feature Suppression)

Specificity tests confirm that the causal path tracing correctly identifies load-bearing features by testing whether high-activation features *outside* the backbone affect the prediction. For each circuit, we suppressed 2 non-backbone features — nodes with high activation or influence scores that do not appear on any of the top-10 causal paths.

Circuit	Feature	Neuronpedia Label	Steered Token	Disrupted?
<code>analog_berlin</code>	L6/3335	"difficulty/challenges"	Germany	no
<code>analog_berlin</code>	L13/4435	"opera-related terms"	Germany	no
<code>analog_rome</code>	L6/2267	"formal text/code" (§3.4)	Italy	no
<code>analog_rome</code>	L4/14857	"code snippets" (§3.4)	Italy	no
<code>analog_tokyo</code>	L6/2267	"formal text/code" (§3.4)	Japan	no
<code>analog_tokyo</code>	L3/10018	early structural feature	Japan	no
<code>analog_teacher</code>	L4/14857	"code snippets" (§3.4)	school	no
<code>analog_teacher</code>	L8/13766	"analogies or comparisons" (§3.3)	school	no
<code>analog_bird</code>	L6/2267	"formal text/code" (§3.4)	sky	YES
<code>analog_bird</code>	L5/5793	"analogies" (§3.3)	air	no
Cairo→Kenya	L5/5500	"profanity and comparisons"	Kenya	no
Cairo→Kenya	L0/8	"are/research"	Kenya	no
Puppy→cat	L9/2909	"formulas/ratios"	cat	no

12/13 pass specificity. The sole exception is instructive: suppressing L6/2267 ("words in programming code, legal jargon, or scientific texts") in the `analog_bird` circuit changes "air" to "sky." Both are semantically valid completions — the model's choice between "air" (p=0.117) and "sky" (likely the next-ranked competitor) is barely resolved, and this formal-text feature tips the balance. Critically, L6/2267 is confirmed **causally inert for all high-confidence circuits** — it does not affect Berlin, Rome, Tokyo, or Cairo. This supports the interpretation from §4.2: formal-text features process template syntax in Phase 1 and do not drive the relational completion in Phase 3, except at the extreme margin where token competition is unresolved.

The L5/5793 ("analogies") feature — the paper's "smoking gun" analogy-concept feature — passes specificity for the bird circuit (prediction unchanged at "air"), consistent with the individual necessity result (§3.7.3) where it is never necessary. This feature is part of the Phase 2 collective that is necessary when suppressed together (§3.7.4) but is individually dispensable in all tested contexts.

3.7.7 Summary of Causal Validation

	Circuit	p	Type	Individual Necessity	Full Suppress	Phase 2 Collective	Sufficiency	Specificity
	analog_berlin	0.973	Capital	1/9	DISRUPTED	→ France	1/2	PASS (0/2)
	analog_rome	0.974	Capital	4/10	DISRUPTED	→ France	1/2	PASS (0/2)
	analog_tokyo	0.990	Capital	3/10	DISRUPTED	→ France	1/2	PASS (0/2)
	analog_teacher	0.486	Sem. role	3/10	DISRUPTED	→ be	0/2	PASS (0/2)
	analog_bird	0.117	Sem. role	8/10	DISRUPTED	→ fish	1/2	1/2 (air→sky)
	Cairo→Kenya	0.963	Capital	2/9	DISRUPTED	—	1/2	PASS (0/2)
	Puppy→cat	0.756	Sem. role	0/4	intact	—	—	PASS (0/1)

Across 159 steering experiments on 7 circuits, the causal validation establishes five principal findings:

1. **The late-layer backbone is collectively necessary.** Full backbone suppression disrupts 6/7 circuits across both geographic and semantic role analogies, confirming that the features identified by causal path tracing drive the prediction.
2. **Individual necessity scales with circuit fragility.** High-confidence circuits have 1–4 necessary features; the lowest-confidence circuit has 8/10. Redundancy through parallel causal paths is a property of well-learned associations.
3. **Phase 2 is collectively necessary but individually redundant.** No individual Phase 2 feature disrupts any high-confidence circuit, but simultaneous suppression of all four Phase 2 features disrupts every circuit — capital analogies revert to the source-pair answer ("France"), demonstrating that Phase 2 implements relational transfer.
4. **Phase 1 template features are individually necessary.** [L0/11651](#) ("the word 'to'") alone disrupts 4/5 circuits, with capital analogies repeating the city name. Template parsing at the earliest layers is a prerequisite for relational reasoning at later layers.
5. **Formal-text features are causally inert.** The high-recurrence features at L4–L6 (§3.4) — [L4/14857](#) "code snippets", [L6/2267](#) "formal text/code" — do not affect any high-confidence prediction when suppressed, confirming they process template syntax rather than driving entity selection. The sole exception (bird: air→sky at p=0.117) occurs at the margin of unresolved token competition.

4. Discussion

4.1 The Analogical Reasoning Circuit in Gemma-2-2B

Our analysis reveals that GEMMA-2-2B implements analogical reasoning through a distributed circuit spanning all 26 transformer layers, with specific functional specialization. The most significant finding

is the existence of **explicitly semantic analogy features** at layers 5, 8, 9, and 13 — features that Neuronpedia's automated explanation system describes using the words "analogies," "comparisons," and "relationships between concepts." The presence of these features suggests that the model has internalized the concept of analogical structure as a discrete, reusable computational primitive, rather than computing analogies purely through implicit pattern matching.

4.2 The Role of Formal Text Features

The high-recurrence "code and legal text" features ([L4 #14857](#) , [L6 #2267](#) , [L3 #3205](#)) present an intriguing puzzle. We propose two complementary interpretations:

- **Functional hypothesis:** These features detect formal, template-driven text patterns. The analogy syntax "X is to Y as Z is to" is highly formal and structured, resembling legal definitions, code comments, and mathematical notation. The model reuses a general "formal syntax" detector.
- **Training data hypothesis:** The analogy format "A is to B as C is to D" appears frequently in SAT prep, vocabulary building, and educational texts — which also contain code examples and legal definitions, creating a statistical association between formal-text features and analogy-completion contexts.

Both are compatible with the data. The co-occurrence of formal-text features (even higher appearance counts) with true analogy features (lower counts, higher specificity) supports interpretation 1: the formal features process the syntax while the analogy features process the relational semantics.

4.3 Cross-Domain Generalization

The consistent activation of the same core features — especially [L5 SAE#5793](#) "analogies" and [L8 SAE#13766](#) "analogies or comparisons" — across both capital-city and semantic-role analogy types provides evidence for a **domain-general analogical reasoning mechanism**. The slightly larger graphs for semantic-role analogies (teacher, bird) may reflect that these completions require broader world knowledge access — knowing that teachers work in schools, or that birds inhabit air/sky — rather than purely relational computation.

4.4 Comparison with the Capital City Recall Circuit

Comparing with the capital city *factual recall* circuit (prompt: "The capital of X is") reveals:

- **Overlap:** [L4 #14857](#) ("code snippets") and [L6 #2267](#) appear with high frequency in *both* circuits — these features are activated by formal definitional prompts generally, not specific to either task.
- **Divergence:** The L5 "analogies" feature and L8 "analogies or comparisons" feature appear to be *specific to the analogical task* — they were not among the top recurring features in the factual recall circuit. This supports the interpretation that these features are selectively activated by relational structure recognition, not by factual recall.

- **Architecture:** Both circuits exhibit layer-spanning computation with early structural processing and later semantic integration, but the analogy circuit shows unique mid-layer (L8–L9) analogy-detector features absent from the factual recall circuit.

5. Limitations

1. **SAE-feature-level intervention only.** Steering operates at the SAE feature level, not the attention head or residual stream level. The causal role of non-SAE circuit components (attention patterns, MLP neurons, residual stream directions) is not assessed. Additionally, while we tested individual and collective phase suppression, combinatorial interactions between specific feature subsets were not exhaustively explored.
2. **SAE coverage.** The `gemmascope-transcoder-16k` SAE covers only cross-layer transcoder features. Attention head contributions and residual stream features are not captured in this analysis.
3. **Threshold sensitivity.** Results are sensitive to the node and edge thresholds set during graph generation (0.80 and 0.85). Lower thresholds would reveal more features; higher thresholds would produce sparser, more focused circuits.
4. **Label quality.** Neuronpedia's automated feature explanations are LLM-generated and may not perfectly capture feature semantics. The "code snippets" label for `L4 #14857` may underspecify a broader "formal structured text" concept.
5. **Prompt set size.** Five prompts are sufficient for initial circuit identification but too few to claim statistical robustness. A larger prompt set covering more analogy types would strengthen conclusions.

6. Conclusions

We have identified and characterized the **analogical reasoning circuit in GEMMA-2-2B** using SAE attribution graphs from the Neuronpedia platform. The key conclusions are:

1. **A stable shared circuit exists.** 180 features are active across all five tested analogy prompts, and 510 across at least three, demonstrating a consistent internal mechanism for analogy completion.
2. **Dedicated analogy features exist at layers 5, 8, 9, and 13.** These features have Neuronpedia explanations explicitly referencing analogies, comparisons, and relational concepts — providing the first direct evidence of interpretable analogy-concept features in a large language model.
3. **The circuit exhibits a three-phase architecture.** Template parsing (L0–L4), analogy recognition (L5–L9), and relational integration (L10–L13), with activation magnitude increasing through the sequence.

4. **Cross-domain generalization is confirmed.** The same core features, including the literal "analogies" feature ([L5 SAE#5793](#)), activate for both geographic and semantic role analogies — suggesting a domain-agnostic relational reasoning primitive.
5. **Formal-text features co-activate.** High-recurrence features encoding "code" and "legal text" patterns reflect the formal syntax of the analogy template activating domain-general structured-text detectors.
6. **Causal steering confirms backbone necessity across all five prompts.** Feature steering experiments (§3.7) on all five prompts demonstrate that full backbone suppression collapses the prediction in every case. Individual necessity scales inversely with confidence: the high-confidence capital analogies have 1–4 necessary features (extensive redundancy), while the low-confidence [analog_bird](#) circuit has 8/10 (nearly every feature is a single point of failure). The formal-text features (§3.4) are confirmed causally inert for 6/7 tested circuits.

Supplementary Materials

Interactive Presentation

20-slide reveal.js presentation with GitHub dark theme, circuit flow diagrams, feature tables, and layer-by-layer analysis.

 [View Presentation](#)

Live Attribution Graphs

Interactive Neuronpedia attribution graph viewer for all five prompts:

[analog_berlin](#) — Paris:France::Berlin?
[analog_rome](#) — Paris:France::Rome?
[analog_tokyo](#) — Paris:France::Tokyo?
[analog_teacher](#) — Doctor:hospital::teacher:?
[analog_bird](#) — Fish:water::bird:?

BibTeX

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Website template based on the [Nerfies](#) and [LatentQA](#) project pages. Analysis performed using Neuronpedia API, NetworkX, and custom Python tooling. Model: GEMMA-2-2B (Google DeepMind).

SAE: [gemmascope-transcoder-16k](#) .